

Aswanth Manoharan

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Summary

Data focused software engineer with hands-on experience building **data-driven web applications, dashboards, and backend systems**. Strong proficiency in **Python, SQL, Django, and React**, with experience integrating multiple data sources, performing analysis, and presenting insights through interactive visualisations. Experienced working with real-world datasets, ETL pipelines, and cloud-based systems, with a strong focus on **data quality, usability, and end-user accessibility**.

Experience

Python Backend Developer (Part-Time) - HitoAI, Dublin Jun 2024 – May 2025

- Designed and maintained **data-driven FastAPI services** processing 10,000+ daily requests.
- Built **data processing and analytics workflows** using Pandas, reducing data visualisation latency by 35%.
- Integrated multiple external APIs and data sources, performing **data validation, transformation, and aggregation**.
- Developed backend systems supporting **dashboard-style data consumption** and reporting use cases.
- Collaborated with frontend teams using **React** to deliver user-facing data visualisation features.
- Documented data models, APIs, and workflows to support team understanding and end-user usage.

Junior Python Developer - Tfora Technologies Mar 2023 – Apr 2024

- Built and maintained **Django REST APIs** supporting structured data exchange.
- Worked with **relational databases (SQL)** for data storage, querying, and reporting.
- Performed **data validation and transformation** ensuring consistency across application workflows.
- Collaborated with cross-functional teams to translate business requirements into backend and data solutions.

Machine Learning Intern - Ignitus Sep 2022 – Feb 2023

- Processed and analysed **50,000+ record datasets** using Pandas and NumPy.
- Applied machine learning techniques (SVM, logistic regression), improving model accuracy by 18%.
- Built reproducible ML pipelines and performed feature engineering and preprocessing.

Projects

ETL Data Warehouse and BI Dashboard System 2024

SQL Server, SSIS, Tableau, Excel

- Designed a **data warehouse (star schema)** to support analytical reporting.
- Built automated **ETL pipelines** processing 65,000+ records.
- Developed **interactive dashboards** visualising trends, performance, and business metrics.
- Performed data cleansing, transformation, and validation ensuring high-quality outputs.

E-Commerce Platform (Django + React) 2024

- Built a **full-stack web application** using Django and React.
- Developed APIs and frontend components for data display, filtering, and user interaction.
- Designed database schema supporting structured data workflows and queries.
- Deployed on cloud infrastructure and maintained application stability.

Real-Time Fall Detection System 2025

- Built a real-time ML system using PyTorch and OpenCV achieving 98.9% recall.
- Developed data pipelines for real-time video processing and analysis.

Skills

Programming: Python, SQL, JavaScript

Web Development: Django, FastAPI, React, REST APIs

Data Analysis: Pandas, NumPy, Data Cleaning, Data Validation

Visualization: Tableau, Matplotlib, Seaborn

Data Engineering: ETL Pipelines, Data Warehousing, SSIS

Databases: PostgreSQL, SQL Server

Tools: Git, GitHub, Docker, AWS (EC2, S3)

Core Strengths: Data Interpretation, Dashboard Development, Problem Solving, Communication, Collaboration

Education

MSc in Data Analytics - Dublin Business School 2024 – 2025

BSc in Chemistry & Industrial Chemistry - University of Kerala 2017 – 2020